VATSIM MINNEAPOLIS ARTCC – MINNEAPOLIS TRACON

LETTER OF AGREEMENT

EFFECTIVE: March 31, 2026

SUBJECT: TERMINAL AREA CONTROL SERVICE

1. PURPOSE. This agreement delegates authority and establishes procedures for the control of IFR and Special VFR traffic on the VATSIM network within the terminal area described herein and is supplementary to FAA Order 7110.65, Air Traffic Control Handbook and VATSIM Network Policies and Procedures.

2. SCOPE. Minneapolis ARTCC (ZMP) delegates to Minneapolis Approach Control (M98) the airspace described in Attachment A for the control of IFR traffic at and below 17,000 ft MSL.

3. WORD MEANINGS. As used in this Letter of Agreement:

a. Altitudes are Mean Sea Level (MSL) and are written in hundreds of feet (110 = 11,000 feet).

b. *Course, Track* and *Heading* are magnetic values.

4. RADAR PROCEDURES. Unless otherwise coordinated, the following procedures shall apply:

a. Handoffs. Aircraft entering ZMP from M98 airspace shall have at least 5 nautical miles (NM) constant or increasing radar separation when radar vectored for the same fix on similar tracks.

b. ZMP releases control for descent and turns ± 30 degrees from assigned heading/course to M98 when aircraft are within 10NM of the ZMP/M98 boundary. M98 must make any necessary point outs to other affected ZMP sectors or RST approach.

c. M98 releases control of departing aircraft to ZMP when the aircraft is at least 25 NM from the MSP DME or leaving 110. ZMP must make any necessary point outs to other affected M98 sectors.

d. M98 may apply procedures specified in FAA Order 7110.65 paragraph 5-5-4 to aircraft transitioning from M98 to ZMP provided that:

(1) The minima are applied only to aircraft on courses/routes that diverge. In-trail aircraft are not included.

(2) M98 maintains communications with at least one of the aircraft until 3-miles lateral separation is attained, and divergence is ensured.

e. Beacon Code Assignments. Normal beacon codes, as assigned by vNAS, shall be used at all times except when vNAS is out of service.

f. Departures

(1) Standard Instrument Departure (SID) routes are depicted in Attachment B.

(2) M98 shall issue enroute clearance provided:

(a) Aircraft are cleared at 170 or below.

(b) Departures filed 060 and below are cleared as filed and assigned an altitude appropriate for direction of flight.

(c) Departures filed 070 or 080 must be on a heading or a course to clear MSP and satellite STAR arrival areas at an altitude appropriate for direction of flight.

(3) M98 must establish departing aircraft filed for 090 above in the appropriate ZMP sector, clear of STAR arrival areas on:

(a) The appropriate MSP/Satellite Airport SID, or

(b) A heading to join the appropriate SID within 60 NM of the MSP DME, or

(c) M98 Satellite departures previously routed via the DWN SID must be routed via INUNE GENEO DKOTA.

(d) For non-SID qualified aircraft, an assigned heading as depicted in Table 1.

TABLE 1. *Non-SID headings to ZMP.*

Sector 10	West of the GEP/BAINY STAR	290° - 300°
	East of the GEP/BAINY STAR	350° - 020°
Sector 09	North of the ENCEE STAR	265° - 285°
Sector 08	South of the TORGY STAR	210° - 240°
Sector 07	East of the KASPR/BLUEM	140° - 160°
RWA	South of the KKILR STAR	100° - 130°
Sector 06	North of the AGUDE STAR	040° - 060°

(4) Prop aircraft requesting 120 or higher must be climbed to or level at 120.

g. Arrivals

(1) MSP and Satellite Standard Terminal Arrival Routes (STARs) are depicted in Attachment C

(2) The clearance limit fix for arrivals shall be the destination airport.

(3) RNAV STAR Procedures:

(a) MSP RNAV STARs and M98 airspace are depicted in Attachments D and D-1 through D-11.

(b) ZMP "Descend Via" phraseology must include a runway transition. ZMP will normally assign runway transitions as depicted in Table 2.

TABLE 2. *Runway Transitions assigned by ZMP.*

	BAINY	MUSCL	KKILR	NITZR	BLUEM	TORGY
12	12L	12L	12R	12R	12R	12R
12/17	12L	12L	12R	12R	12R	12R
30	30R	30R	30L	30L	30L	30L
30/17	30R	30R	30L	30L	30L	30L
30/35	30R	30R	30L	30L[1]	30L[1]	30L
17/22	17	22	N/A	22	22	17

1 – M98 and ZMP CIC/TMU will collaborate which aircraft will be issued the runway 35 transition.

(c) Heavy Jet arrivals must be assigned the Runway 12R/30L transition.

(d) Transfer of communications of aircraft entering M98 airspace on an RNAV STAR must be accomplished no later than the following fixes:

STAR	TORGY	BAINY	MUSCL	KKILR	BLUEM	NITZR
FIX	OFSON	LUCCY	BAYKS	KKILR	BLUEM	NITZR

(c) ZMP must verbally coordinate any turbojet aircraft cleared via a conventional STAR with the appropriate M98 Sector and cross the boundary at LOA altitudes and de-conflicted with Optimized Profile Descent (OPD) traffic.

(d) ZMP must verbally coordinate any assigned speed that deviates from those published on the RNAV STAR, except those speeds that are part of a TMU initiative.

(e) When MSP is on Runway 30/17 configuration, Knock It Off procedures will be implemented on the NITZR and BLUEM arrivals. No "Descend Via" instructions will be issued. Runway transitions will be issued. Any speed other than published speed must be coordinated.

1 NITZR Arrival X WRSAW at and maintain 110

2 BLUEM Arrival X HHAMR at and maintain 100

NOTE - *M98 will verify assigned altitude.*

(f) ZMP Sector 7 must provide M98 with a minimum of 5 miles in trail between arrivals on the same STAR, and 7 miles staggered spacing between arrivals on parallel STARs.

(g) "Knock It Off" OPD Procedures. When MSP is landing Runway 4, 22, 17, 17/22, 22 or anytime either facility has an operational necessity to discontinue the OPD portion of the STAR, ZMP must issue crossings as depicted in Table 3.

TABLE 3. *Knock It Off Crossing altitudes*

MUSCL Arrival	Cross BAYKS at and maintain 120
KKILR Arrival	Cross HUGGI at and maintain 100
BLUEM Arrival	Cross HHAMR at and maintain 100
NITZR Arrival	Cross WRSAW at and maintain 110
TORGY Arrival	Cross OFSON at and maintain 110
BAINY Arrival	Cross LUCCY at and maintain 110

1 Cross the M98 boundary at the published speed of 280kts unless otherwise coordinated.

2 Coordination must be accomplished through M98 CIC/TMU and ZMP CIC/TMU.

3 ZMP/M98 may invoke "Knock It Off" OPD procedures at individual gates regardless of runway configuration when operational necessity dictates.

(4) Non-RNAV Arrival Procedures. To the extent practical, all turbojet and turboprop aircraft must be routed via STARs.

(a) Turbojet aircraft that will not be on a conventional STAR for equipment or route of flight reasons must be coordinated. These aircraft must be on a heading through an arrival gate to approximate direct MSP airport and at the appropriate "knock it off" altitude for the gate.

(b) Turboprop aircraft that will not be on a conventional STAR for equipment or route of flight reasons must be coordinated. These aircraft must be routed direct MSP through an arrival gate, or on a heading through an arrival gate to approximate direct MSP at 070.

(5) All turbojet and turboprop aircraft must be routed via STAR routings. The appropriate crossing altitude restriction must be issued as depicted in Table 4.

TABLE 4. *Conventional STAR Crossing Altitudes.*

	Aircraft	STAR	Crossing Restriction
MSP Arrivals	Turbojets	GEP KASPR	OLLEE at 110 DELZY at 100
	Turboprops	GEP	Landing 12s OLLEE at 70[2] Landing 30s or 30/35 OLLEE at 90
Satellite Arrivals	Turbojets	AGUDE ENCEE GEP TWOLF	AGUDE at 80 RIXIE at 80 OLLEE at 70[1] TRGET at 70
	Turboprops	AGUDE ENCEE GEP TWOLF	AGUDE at 60 RIXIE at 70 OLLEE at 70[2] TRGET at 70

1 – 6,000 ft. when MSP is landing Runway 17.
2 – 5,000 ft. when MSP is landing Runway 17.

(6) STP, ANE, and FCM arrivals routed over BITLR must be assigned direct GEP and cross BITLR at 060.

(7) FCM and LVN arrivals entering M98 south of RGK may be routed direct FGT direct destination without coordination at the following altitudes:

(a) MSP landing 30's, 30/17, 30/35 at 040.

(b) Any other runway configuration at 040 or 060.

(8) FCM and LVN arrivals from ZMP Sectors 08 and 09 may be cleared direct destination at 5,000 ft. Aircraft may still be cleared via the TWOLF Arrival or the ENCEE Arrival.

(9) FCM arrivals from ZMP sector 10 may be issued direct to FCM west of the GEP arrival gate at or below 050.

(10) All piston engine aircraft terminating in M98 airspace must enter M98 Satellite Airspace at or below 060 at an altitude appropriate for direction of flight.

(11) STC departures landing MSP and M98 satellite airports must enter M98 airspace at 050. When Runways other than 12L/R are in use, turbojet and turboprop aircraft departing STC and landing MSP must be on a STAR arrival at the appropriate altitude

(12) MKT departures landing M98 satellite airports must enter M98 airspace cleared as filed at 050.

(13) ZMP must have control for descent to 100 on aircraft exiting M98 airspace and landing RST Airport when the aircraft are 25 NM from the MSP DME.

(14) Aircraft landing OEO, RNH, or RGK from the east (through Sector 5) must be level at, or descending to 040. M98 must have control when aircraft are within 10 NM of the M98 boundary.

(15) Aircraft landing STC or MKT will enter ZMP airspace at or descending to 040. ZMP has control on contact.

(16) During time periods outside of normal operating hours in which ZMP assumes Rochester Approach Control (RST) airspace, unless otherwise coordinated, all turboprop aircraft routed via the KASPR STAR must cross the DELZY intersection at 070, and all piston engine aircraft terminating in M98 airspace must exit RST airspace at 060.

(17) Frequencies. ZMP must assign frequencies to arrivals/overflights as depicted in Attachments D-1 through D-11.

(5) **RUNWAY 17-22 LANDING CONFIGURATION PROCEDURES.** When the Minneapolis-St. Paul Airport is in a runway 17/22 landing configuration, unless otherwise coordinated, the following procedures must apply:

a. Departures. All existing departure routes and procedures remain unchanged except:

(1) Jet aircraft filed for the WLSTN SID must be vectored north of the AGUDE gate and assigned a heading to join the WLSTN SID. Normal altitudes apply.

(2) Turboprop aircraft filed for the WLSTN SID or over EAU, 7,000 MSL or higher, must be cleared via the COULT SID to the COULT intersection, direct EAU, direct next flight planned fix.

(3) Jet and Turboprop aircraft filed to, or over, DLH, EVM, RZN, and HIB must be vectored west of the AGUDE gate to enter Sector 6 established on an assigned heading between 020 and 040 degrees. Normal altitudes apply.

(4) Jet and Turboprop aircraft filed to, or over, BRD must enter Sector 10 west of the OLLEE intersection established on a heading between 290 and 310 degrees. Normal altitudes apply.

(5) STP, ANE, and MIC north bound departures are exempt from the above restrictions and must exit M98 airspace established on a heading between 350 and 010 degrees.

(6) Departures filed at 5,000 or 6,000 feet MSL must be on a heading to clear the GOPHER and AGUDE satellite STAR arrival areas.

(7) M98 may clear KBREW SID traffic direct to KBREW from positions south of the SID.

b. MSP Arrivals. Knock It Off Procedures are in effect for MSP arrivals.

(1) Turboprops cross OLLEE at 5,000 feet MSL.

(2) Turboprops cross TWINZ at 6,000 feet MSL.

(3) KKILR/WILDD Arrivals: All KKILR/WILDD arrivals are prohibited.

c. Satellite Arrivals.

(1) GOPHER Satellite Arrivals:

(a) Jets cross OLLEE at 6,000 feet MSL.

(b) Turboprops cross OLLEE at 5,000 feet MSL.

(c) M98 must have control for descent of GEP STAR satellite traffic within 10NM of the M98/ZMP airspace boundary.

(2) AGUDE Satellite Arrivals:

(a) STP traffic must be assigned the AGUDE STAR.

(b) FCM traffic normally routed over AGUDE must be routed over BITLR and assigned direct FGT direct FCM and cross BITLR at 6,000 feet MSL. APREQ is not necessary.

(c) Jets cross AGUDE at 6,000 feet MSL.

(d) Turboprops cross AGUDE at 5,000 feet MSL.

(e) Stacks at AGUDE for aircraft landing the same airport are not authorized.

(f) M98 must have control for descent of AGUDE traffic within 10NM of the M98/ZMP airspace boundary.

d. Overflights. The triangular airspace north of the MSP airport between and including Runways 17 and 22 ACDAs must be avoided from all directions below 13,000 feet MSL.

5. MIDNIGHT OPERATIONS. When coordinated (between the hours of 2230 and 0530 local):

a. ZMP shall clear all traffic landing at airports within M98 as follows:

(1) Direct to the destination airport.

(2) Via pilot discretion descent to 10,000 feet to all aircraft at altitudes at and above 10,000 feet.

(3) Arrivals at and below 10,000 feet shall remain at altitude.

(4) ZMP shall assign all aircraft entering M98 airspace frequency 124.7 (MSP_R_APP)

b. M98 shall:

(1) Clear departure aircraft via the first fix in the route.

(2) Point out departures to any affected Center position.

(3) Climb turboprops/props to 17,000 feet, or lower altitude if requested.

6. ATTACHMENTS

Attachment "A" — ZMP Area/Sector Map and Frequencies.
Attachment "B" — SIDs.
Attachment “C” — Conventional STARs
Attachment “D” — MSP RNAV STARs
Attachment D1-D11 — Approach Control Airspace and Frequencies

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ATTACHMENT “A”

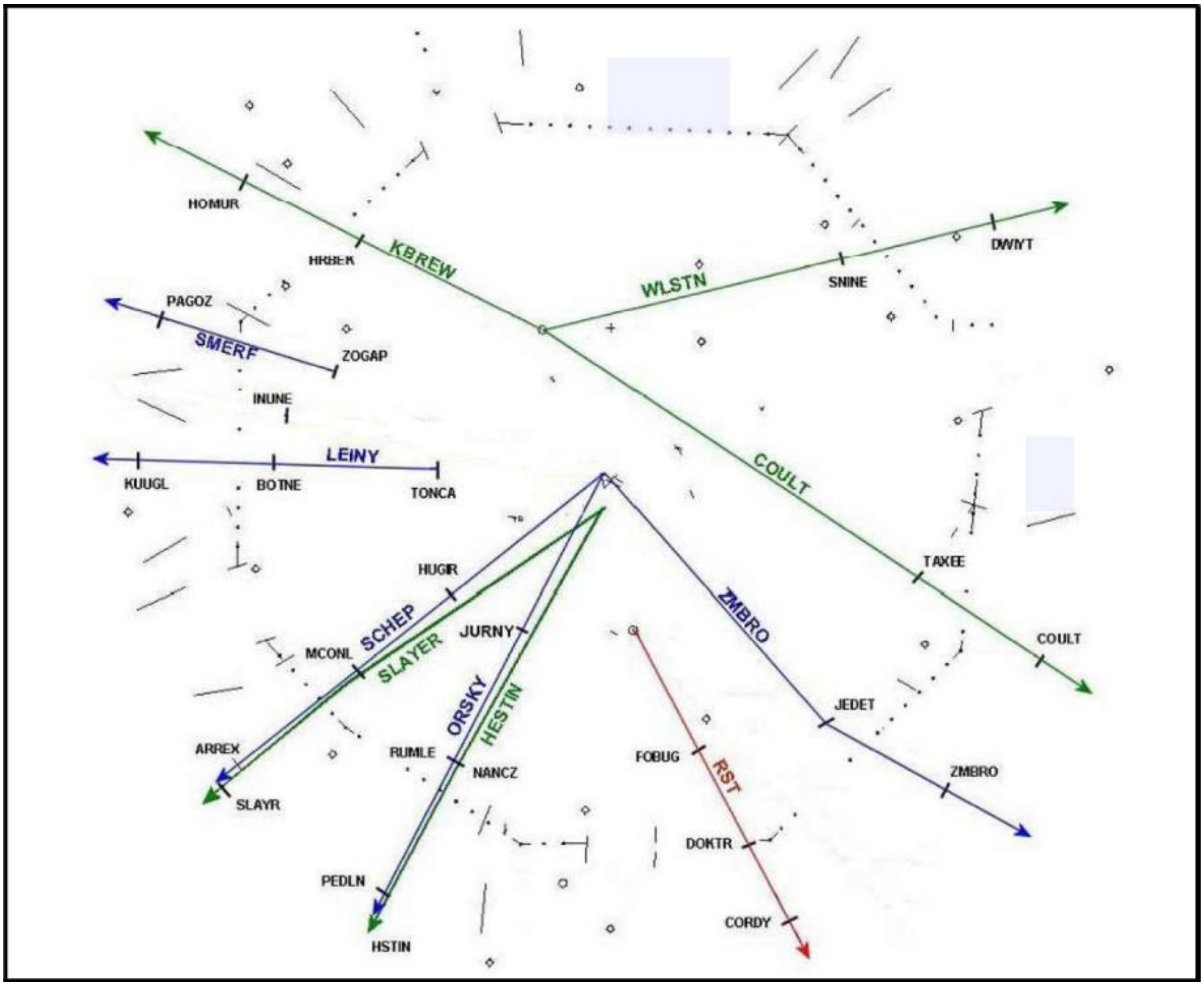
ZMP Area/Sector Map and Frequencies

ZMP FREQUENCIES

(05)125.3/335.6(EAU)
128.6/363.0(LSE)
(06) 134.3/285.5(ZMP)
(07)132.35/307.3(RST)
(08)135.0/306.9(MKT)
(09)125.5/323.1(DWN)
(10)121.05/397.9(PNM)
(RWA)134.85/352.0(BKB)

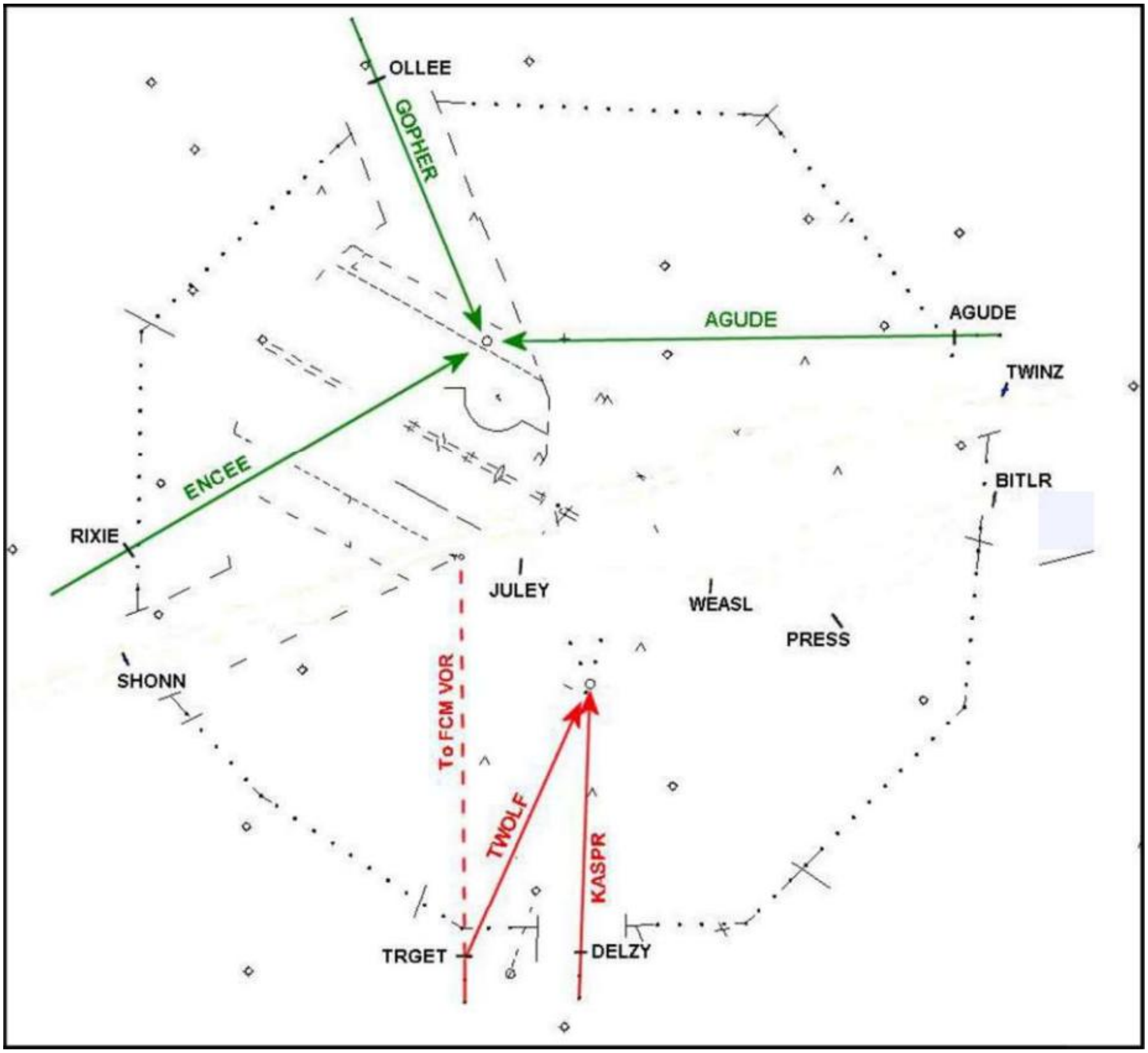
ATTACHMENT “B”

SIDs



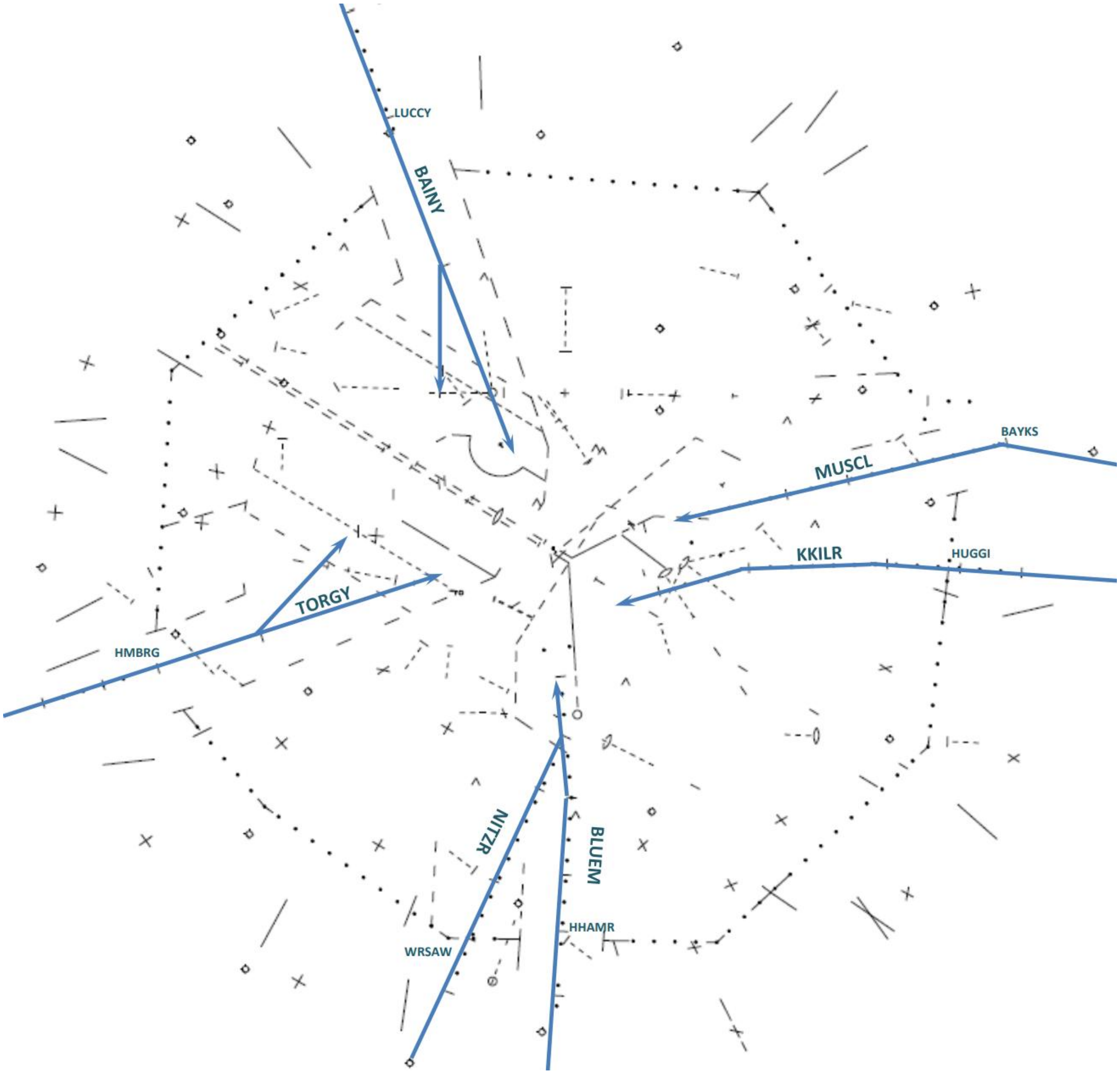
ATTACHMENT "C"

Conventional STARs



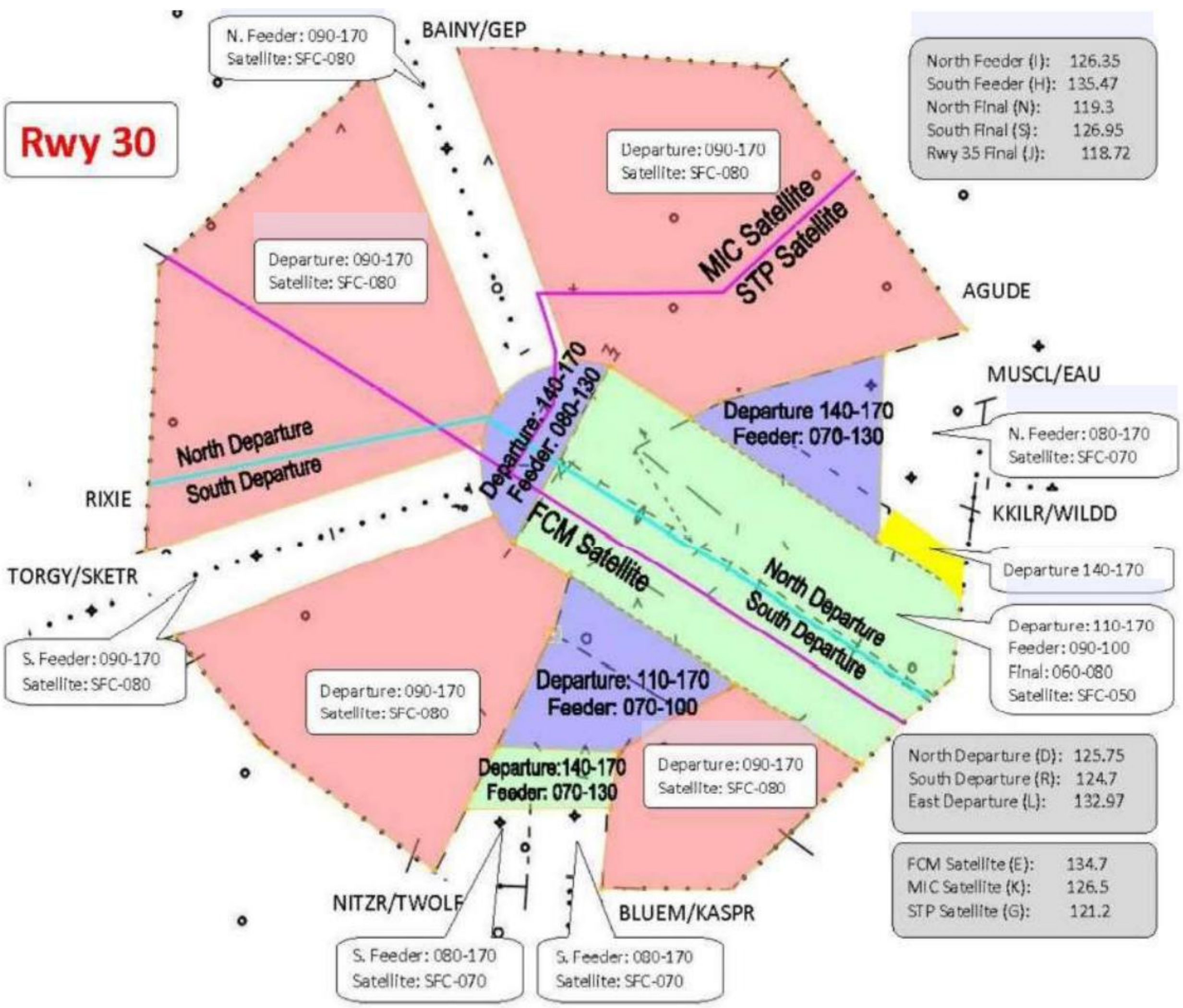
ATTACHMENT "D"

MSP RNAV STARs



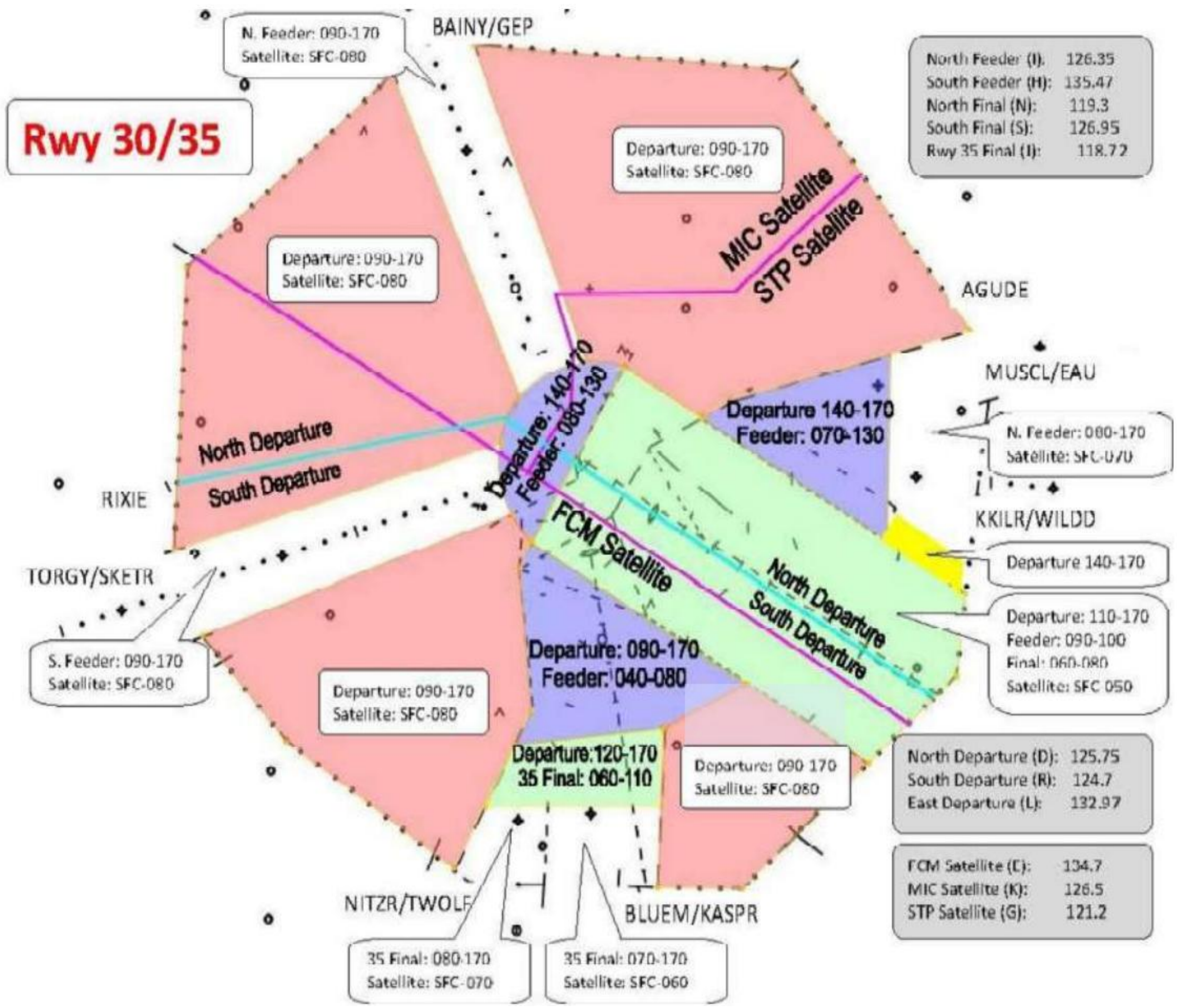
Attachment D-1

Approach Control Airspace and Frequencies



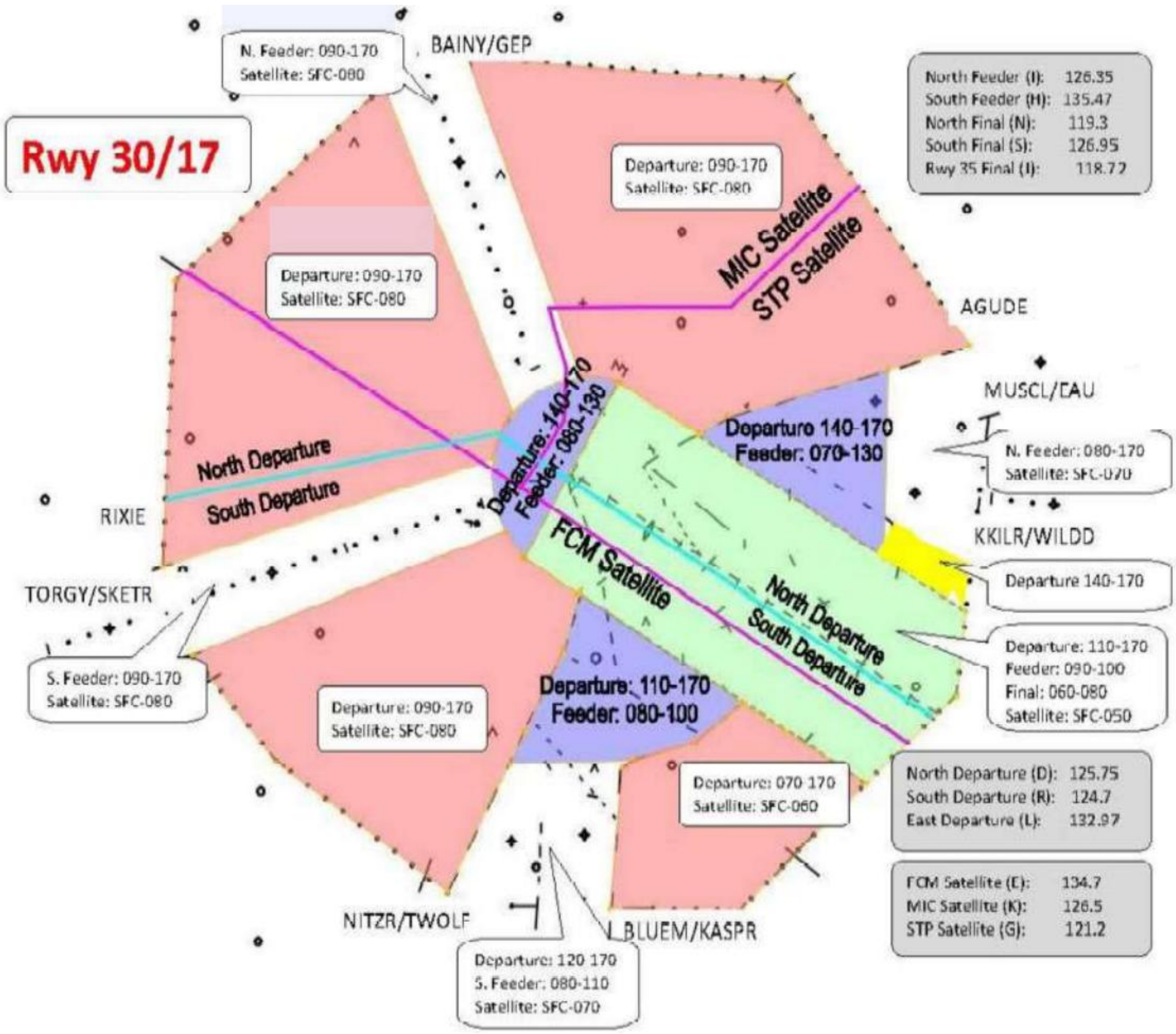
Attachment D-2

Approach Control Airspace and Frequencies



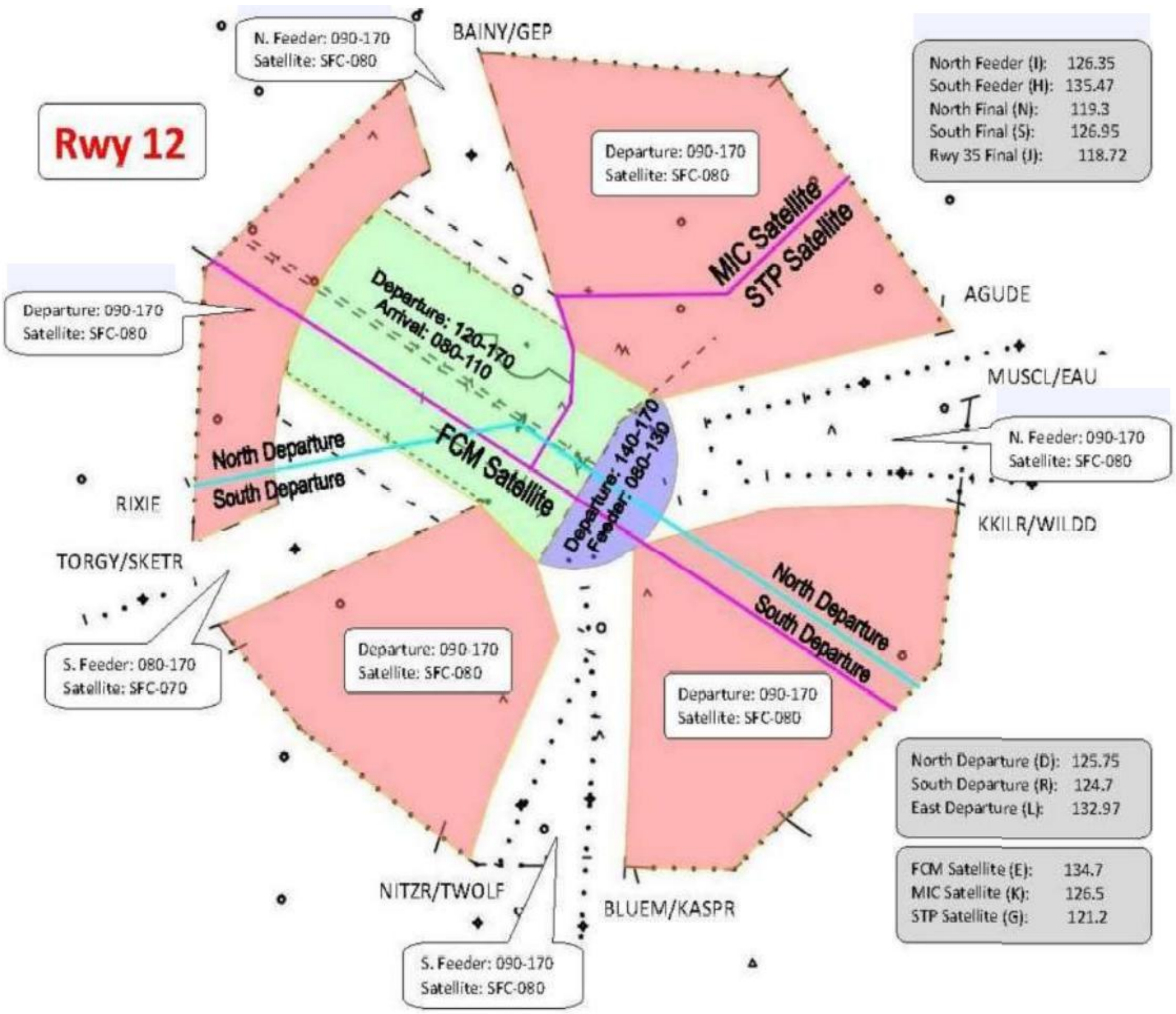
Attachment D-3

Approach Control Airspace and Frequencies



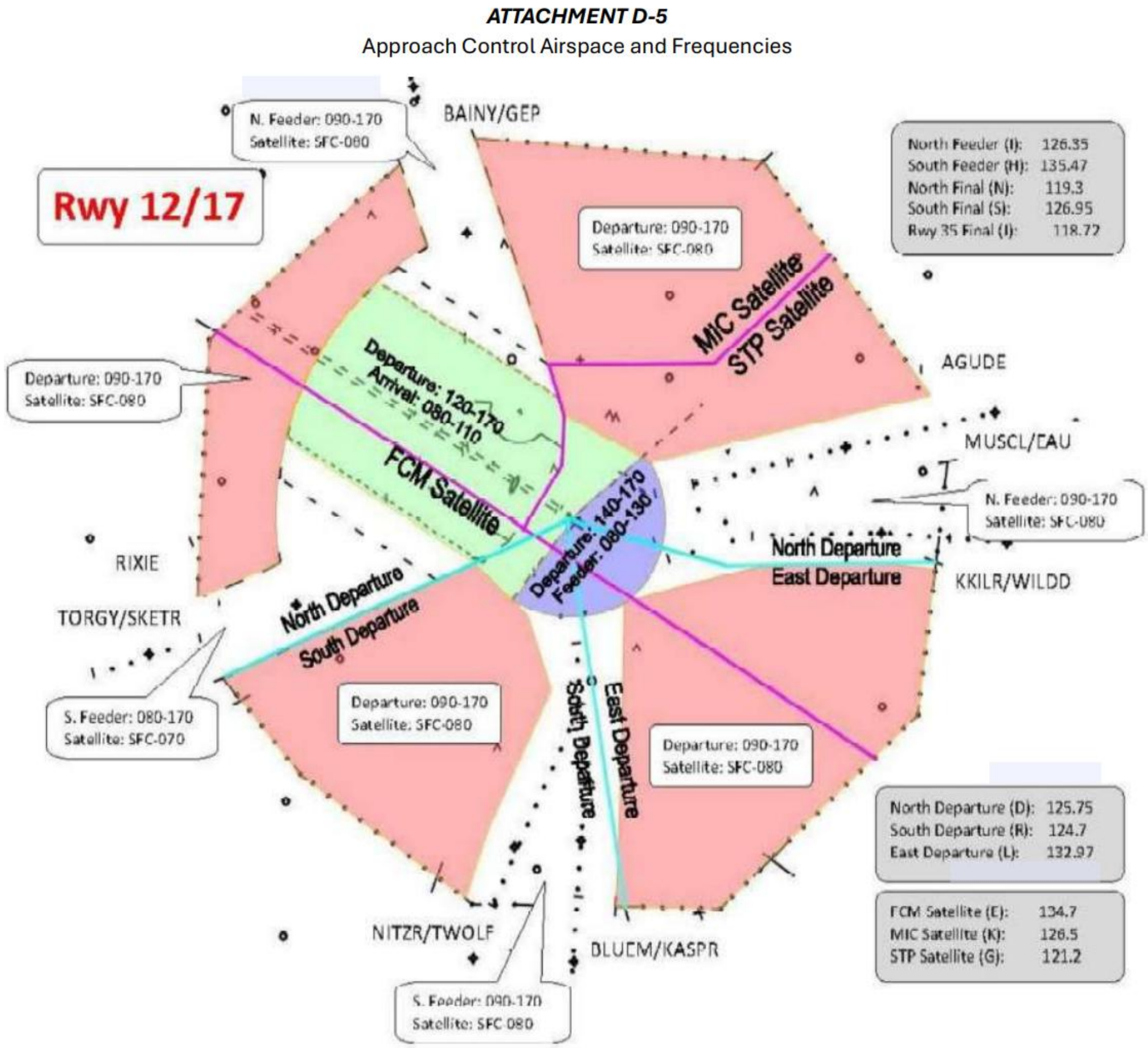
Attachment D-4

Approach Control Airspace and Frequencies



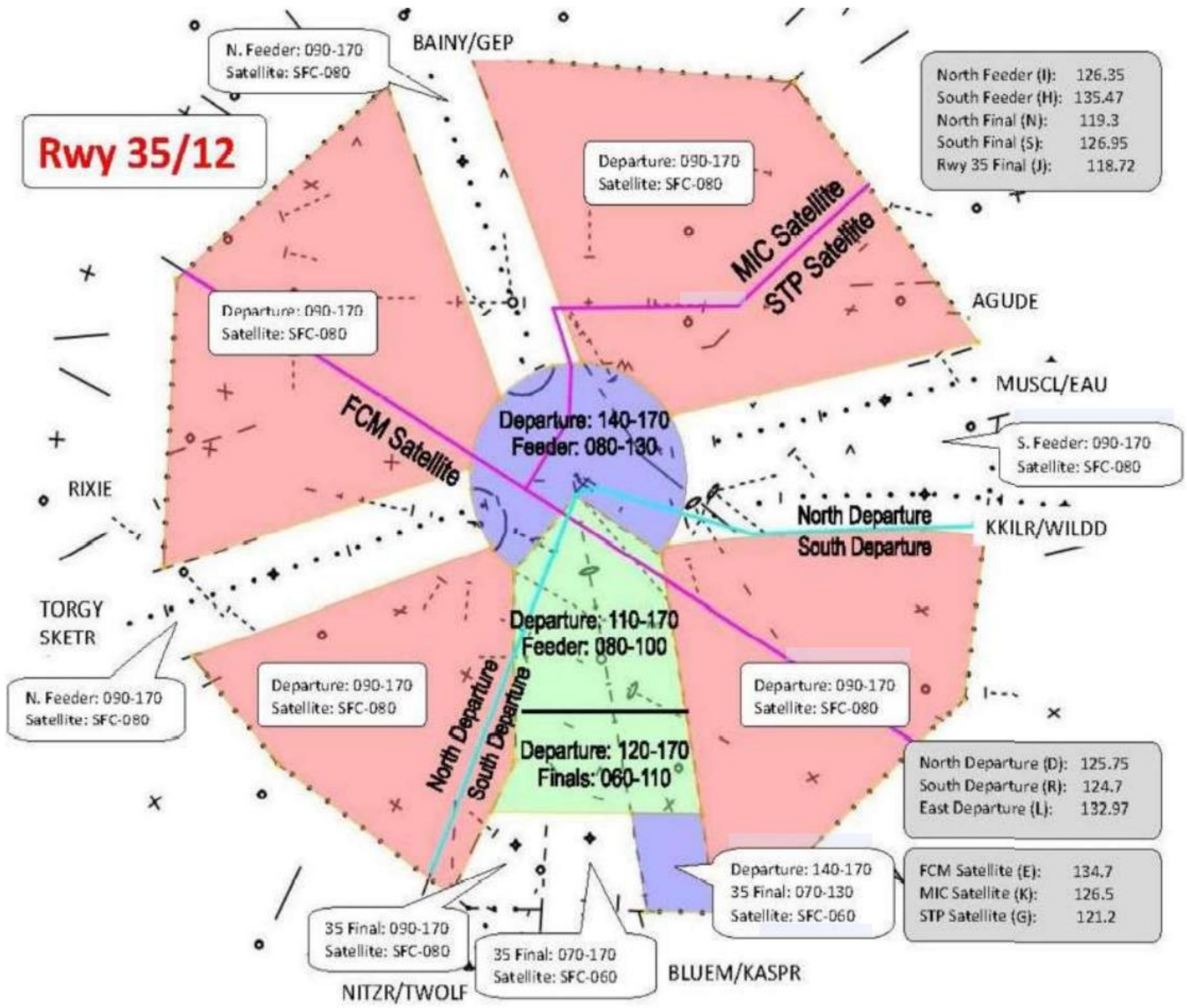
Attachment D-5

Approach Control Airspace and Frequencies



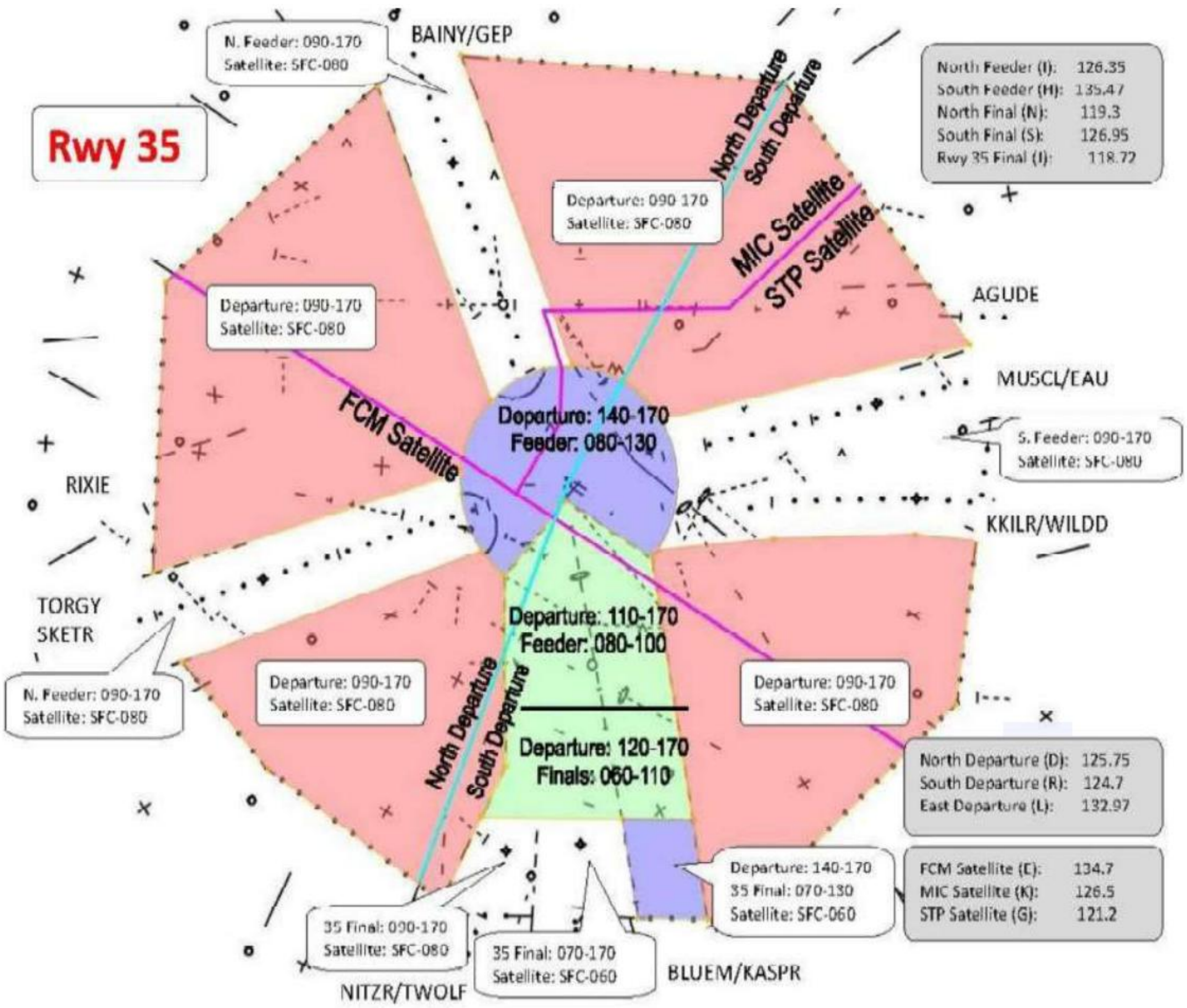
Attachment D-6

Approach Control Airspace and Frequencies



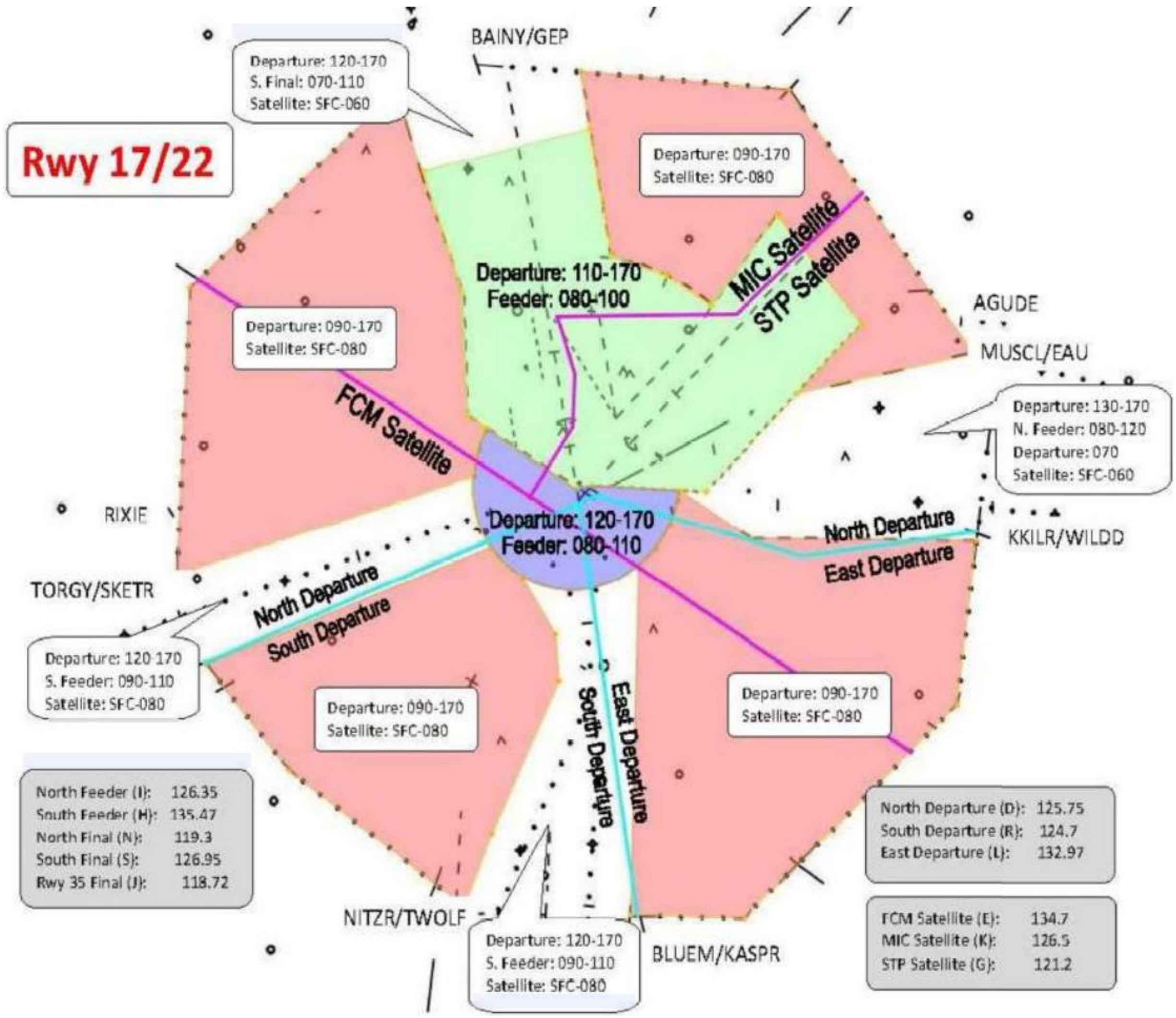
Attachment D-7

Approach Control Airspace and Frequencies



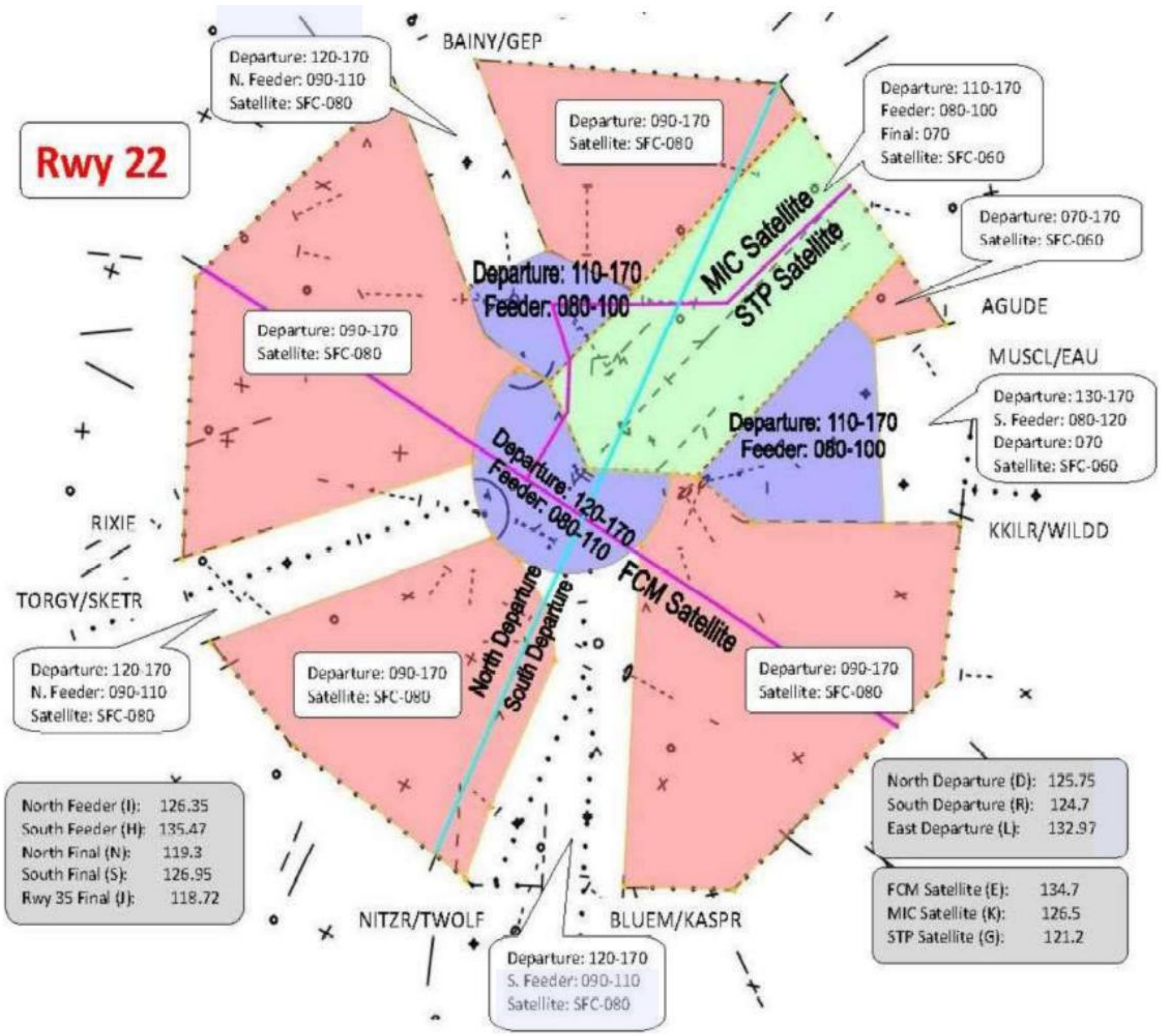
Attachment D-8

Approach Control Airspace and Frequencies



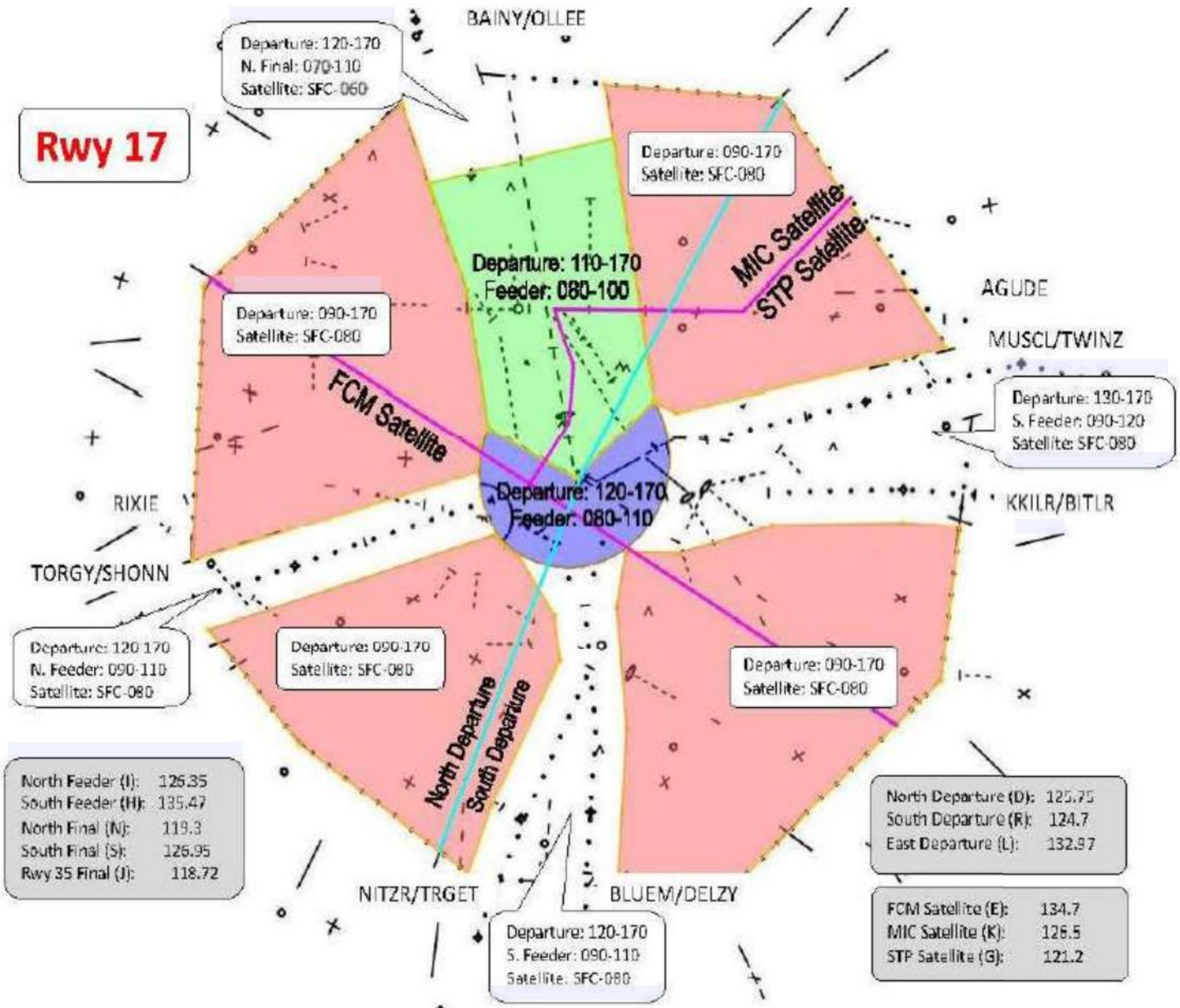
Attachment D-9

Approach Control Airspace and Frequencies



Attachment D-10

Approach Control Airspace and Frequencies



Attachment D-11

Approach Control Airspace and Frequencies